The observed temperature power spectrum through fourth order in a radiation-only universe: final report

Unified analysis (branch `unified/temperature-spectrum`)

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Abstract

We compute the observed temperature anisotropy through third order, and the angular power spectrum through fourth order, in the primordial amplitude ϵ. The setting is a flat radiation-only universe in general relativity, with Planck test photons emitted on $T_{\rm phys} = T_\star$ and an observer on a fluid worldline.

The calculation uses the fluid's velocity potential and comoving labels as coordinates (*temperature frame*). In these coordinates the observed temperature is an exact line integral of the shear of a conformally rescaled metric, and the observer's own temperature cancels exactly in the normalised map. Further results:

- The spatial Einstein equations take a compact mixed trace-free form, which yields a triangular solution scheme with elementary Green functions at every order.
- This scheme satisfies all ten field equations at every order (proved).
- All second-order field kernels are given in closed form for general momenta, in the entire functions Si, Cin and $\mathfrak{e}_m$. This makes every resonance surface manifestly regular.
- Third-order kernels are explicit nested integrals.
- The complete map $\Theta^{(1,2,3)}$ is constructed. It retains photon propagation, emission surface, observer clock, tetrad and normalisation terms.
- $C_\ell^{(11)}$, $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are reduced to exact finite-dimensional integrals, with every sky integral done in closed form.
- The finite-time map admits a compact centred-exponent form, and the spectrum is the sum of a corrected linear-response norm and a quadratic-response norm. These are exact reorganisations through the required order, not evaluations of the remaining integrals.
- In the superhorizon limit the observable is a local trace-free-curvature quadrupole, *exact in* ζ. Its leading-gradient Gaussian spectrum resums to

$$C_{2,\rm lead} = \tfrac{4\pi}{225}\epsilon^2\sigma^2\Xi^2 e^{8\epsilon^2\sigma^2}\big(1+\tfrac{5}{3}\epsilon^2\sigma^2\big), \qquad \Xi \equiv x_o^2 - x_\star^2.$$

  Through fourth order this gives the factor $1 + \frac{29}{3}\epsilon^2\sigma^2$. The resummation concerns the leading-gradient expression, not the finite-wavelength solution.

Analytical proofs and exact computer-algebra checks (no numerical evidence) are distinguished in the verification ledger. We state explicitly which integrals remain unevaluated and which limits remain open.

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1 Problem and summary of results

The task (`problem_statement.tex`) asks for $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ for the observed map

$$\Theta(\hat{n}) = \frac{T_{\rm obs}(\hat{n})}{\frac{1}{4\pi}\int \mathrm{d}\Omega'\, T_{\rm obs}(\hat{n}')} - 1, \qquad T_{\rm obs} = T_\star \frac{E_o}{E_e}, \quad E = -u_\mu p^\mu.$$

The physical model is a perfect radiation fluid with $P = \rho/3$, $\rho = \alpha T_{\rm phys}^4$ and $\Lambda = 0$, with Gaussian adiabatic primordial perturbations and no independent higher-order modes. The calculation must be analytical and keep every term, with all normalisations fixed. A monochromatic specialisation is to be made only at the end.

Requirement	Result (section; status)
Gaussian variable; no higher-order modes	$\zeta = \frac{1}{6}\ln\det\gamma_{ij}$, exactly conserved; Def. 4 (§4)
Constrained initial data at every order	regular solution at any t_i; constraints proved (Thm. 5); forced data nonzero

Requirement	Result (section; status)
Field solution through ϵ^3 incl. vectors/tensors	triangular scheme (§5); 2nd order [closed form]; 3rd order [explicit nested integrals]
Observer, clock, tetrad, emission surface	§6; explicit Fermi–Walker rotation, irrelevant to the spectrum at this order (Prop. 7)
Complete $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$	§6, with an independent generator and Ward checks
$C_\ell^{(11)}$	[closed form] ($\ell = 2, 3$; general ℓ as a 1D integral)
$C_\ell^{(22)}$, $C_\ell^{(13)}$	[exact finite-dimensional integrals], sky integrals evaluated (§8); existence proved
Monochromatic, held-fixed quantity	$\langle g^2 \rangle = \sigma^2$ fixed (§8.2)
Asymptotic limits	superhorizon: [closed form], including an all-amplitude Gaussian average of the leading-gradient map; late time: [first order]; general k_0 at fourth order: [unevaluated integrals] (§9)
Convergence for general spectra	compact support: [proved]; IR: leading cancellation [derived] from the Ward identities; UV: [power counting only]
Verification	§10 (exact symbolic, generic-jet proofs)

2 Conventions and the experiment

Signature $(-,+,+,+)$, $c = 1$. The background is $a = a_1 t$ and $\bar{T} = \mathcal{T}/(a_1 t)$, with $8\pi G\alpha\mathcal{T}^4 = 3a_1^2$.

Fourier and perturbative conventions.

- $g(X) = \int_k g_k e^{ik\cdot X}$, with $\int_k = \int \mathrm{d}^3k/(2\pi)^3$ and $\langle g_k g_{k'} \rangle = (2\pi)^3\delta^3(k + k')P(k)$; also $\mathcal{P}(k) = k^3P/2\pi^2$.
- Every field is $Y = \sum_r \epsilon^r Y^{(r)}/r!$ with $Y^{(r)} = \int_{k_1\ldots k_r} K_r(k_1, \ldots, k_r)\, g_{k_1} \cdots g_{k_r} e^{iK\cdot X}$ and $K = \sum k_a$. The symmetric kernels carry no extra $1/r!$.
- *Multilinear rule:* K_r is the coefficient of $z_1 \cdots z_r$ when $g = \sum_a z_a e^{ik_a\cdot X}$ is inserted and only monomials in distinct labels are kept. This fixes every combinatorial factor.
- Spherical harmonics $Y_{\ell m}$ are complex, with the Condon–Shortley phase.

Measurement labels.

- *Emission:* the surface $T_{\mathrm{phys}} = T_\star$. Emission is isotropic and Planckian in the fluid rest frame. The intersection is the branch continuously connected to the background one.
- *Observer:* the fluid line $X = 0$. The initial event is $T_{\mathrm{phys}} = T_I > T_\star$ on that line, and observation follows after fixed proper time τ_o.
- *Tetrad:* Fermi–Walker transported, with initial orientation $\gamma(t_I, 0)^{-1/2}$ (the symmetric square root, aligned with the marked labels).

- *Background times:* $\bar{t}_\star = \mathcal{T}/(a_1 T_\star)$, $\bar{t}_I = \mathcal{T}/(a_1 T_I)$ and $\bar{t}_o^2 = \bar{t}_I^2 + 2\tau_o/a_1$. We write $x \equiv kt/\sqrt{3} = k/(\sqrt{3}\,aH)$.

3 Exact framework

Lemma 1 (Temperature frame)**.** *For an irrotational radiation fluid, $Tu_\mu = -\mathcal{T}\partial_\mu t$ defines the velocity potential t. In coordinates (t, X^i), with comoving labels X^i,*

$$g = N^2\big(-\mathrm{d}t^2 + \gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j\big), \qquad N = \mathcal{T}/T_{\rm phys}, \qquad u = N^{-1}\partial_t$$

exactly. Euler's equation holds identically, and continuity is equivalent to $\partial_t \det\gamma = 0$. Hence $\gamma = e^{2\zeta(X)}\exp H$ with $\operatorname{tr} H = 0$, and the shear $\sigma_{ij} = \frac{1}{2}\partial_t\gamma_{ij}$ is traceless.

Lemma 2 (Exact redshift)**.** *Null geodesics of g and of $\tilde{g} = g/N^2$ coincide, and $E = \tilde{E}/N$. Hence*

$$T_{\rm obs}(\hat{n}) = T_{\rm fluid}(o)\, e^{-I(\hat{n})}, \qquad I = \int_{t_e(\hat{n})}^{t_o} \sigma_{ij}(t, X(t))\, e^i e^j\, \mathrm{d}t,$$

where $\mathrm{d}X^i/\mathrm{d}t = e^i$, $\gamma_{ij}e^ie^j = 1$ and $\mathrm{d}e^i/\mathrm{d}t = -2S^i{}_j e^j - \Gamma^i{}_{jk}e^je^k + e^i\sigma_{jk}e^je^k$, with $S = \gamma^{-1}\sigma$. The emission event is the root of $\ln t_e + \nu(t_e, X(t_e)) = \ln\bar{t}_\star$, with $N = a_1 t e^\nu$.

$T_{\rm fluid}(o)$ is direction-independent. It therefore cancels from Θ *exactly*. The emission-surface temperature shift, Sachs–Wolfe, Doppler and integrated Sachs–Wolfe terms are all contained in the single integral I.

Proposition 3 (Unified field equations)**.** *With $n = \ln N$, Einstein's equations are equivalent to*

$$(N)\quad 3\nu'' - \Delta_\gamma\nu - 2|D\nu|^2 + \operatorname{tr} S^2 + \tfrac{3}{t^2}(e^{-2\nu} - 1) = 0, \tag{1}$$

$$(T)\quad \mathrm{TF}\left[S' + 2(t^{-1} + \nu')S + R^k{}_j[\gamma] - 2D^kD_j\nu + 2D^k\nu D_j\nu\right] = 0, \tag{2}$$

$$(M)\quad D_jS^j{}_i - 2\partial_i\nu' + \tfrac{2}{t}\partial_i\nu + 2S^j{}_i\partial_j\nu + 2\nu'\partial_i\nu = 0, \tag{3}$$

$$(\mathrm{Ham})\quad R - 4\Delta\nu - 2|D\nu|^2 - \operatorname{tr} S^2 + 6(t^{-1} + \nu')^2 - 6e^{-2\nu}/t^2 = 0. \tag{4}$$

In the mixed form (T), the $\sigma\sigma$ terms of the spatial equation cancel, because $\gamma^{ki}\sigma'_{ij} = (S^k{}_j)' + 2S^k{}_iS^i{}_j$. This is proved for generic γ (six free functions, t-independent determinant) and arbitrary N, on independent symbolic 2-jets.

4 Primordial state and initial data

Definition 4 (Gaussian variable; absence of independent higher-order modes)**.** The state is the solution of Prop. 3 with the following properties:

(G) $\zeta = \epsilon g$ exactly, with g a zero-mean, homogeneous, isotropic Gaussian field;

(T) $\gamma \to e^{2\zeta}\delta_{ij}$ as $t \to 0$. The limit is conformally flat in the marked conformally Cartesian material chart, so there is no tensor seed;

(V) $\mathrm{d}(Tu) = 0$, so there is no vorticity seed;

(R) ν and H are even and analytic in t at fixed momenta, with $\nu(0) = 0$ and $H(0) = 0$, so there are no decaying modes.

"No independent higher-order modes" means $\zeta^{(r\geq 2)} \equiv 0$ together with (T), (V) and (R) at every order.

Because $\det\gamma$ is t-independent, $\zeta = \frac{1}{6}\ln\det\gamma$ at *every* time. As $t\to 0$, constant-t slices become uniform-density slices, so ζ is the nonlinear uniform-density (δN) curvature perturbation.

The residual conformal relabellings are fixed as follows: the dilation by $\langle g\rangle = 0$; special conformal transformations by statistical homogeneity; translations and rotations are statistical symmetries.

Exactly in ζ, the gradient-expansion data are

$$\nu = \tfrac{t^2}{18}e^{-2\zeta}\big(2\nabla^2\zeta + |\nabla\zeta|^2\big) + O(t^4), \quad H_{ij} = \tfrac{t^2}{3}e^{-2\zeta}\ \mathrm{TF}\big[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\big] + O(t^4). \tag{5}$$

The initial data at any t_i are the regular solution of §5 evaluated at t_i, with $\rho = 3a_1^2/(8\pi G N^4)$ and $u = N^{-1}\partial_t$. Higher-order data are nonzero. For example, $\gamma^{(2)}(0) = 4g^2\delta$, and the $\nu^{(2)}$ kernel is $\frac{t^2}{9}[2(k_1^2+k_2^2) - k_1\cdot k_2] + O(t^4)$. The forced $K=0$ mode ($5k^2t^2/9$ at $k_2 = -k_1$) is kept. The data evolve back into the same regular solution, so t_i drops out identically.

5 Field solution at every order

5.1 Triangular scheme

At total momentum K, the order-r kernels (ν_r, A_r), with A_r the order-r part of H, solve $\mathcal{L}_K[\nu_r, A_r] + c^{(r)} = 0$. The operators are the same at every order:

$$\begin{aligned} L_N &= 3\nu'' + K^2\nu - \tfrac{6}{t^2}\nu, \qquad (L_M)_i = \tfrac{i}{2}K_jA'_{ji} - 2iK_i(\nu' - \nu/t), \\ L_T &= \tfrac{1}{2}(A'' + \tfrac{2}{t}A') + \mathrm{TF}\,R_1[A] + 2\,\mathrm{TF}(K_iK_j)\nu. \end{aligned}$$

Here $R_1[A]_{ij} = \frac{1}{2}(K^2A_{ij} - K_aK_iA_{aj} - K_aK_jA_{ai})$. The source $c^{(r)}$ is the multilinear coefficient with $(\nu_r, A_r) = 0$; it is explicit in the lower orders.

Write $c = K/\sqrt{3}$, $x = ct$, $u_1 = \sin x/x - \cos x$ and $u_2 = -\cos x/x - \sin x$. The coefficient c_r is the t^2 coefficient of (5) at order r. In particular $c_2 = [2(k_1^2+k_2^2) - k_1\cdot k_2]/9$ and $c_3 = \frac{2}{9}[-2\sum k_a^2 + \sum_{a<b} k_a\cdot k_b]$. Then

$$\nu_r = \frac{9c_r}{K^2}u_1(x) - \frac{1}{3c}\Big[u_2(x)\int_0^t u_1(cs)c_N^{(r)}\mathrm{d}s - u_1(x)\int_0^t u_2(cs)c_N^{(r)}\mathrm{d}s\Big], \tag{6}$$

$$h_S = \tfrac{3}{2K^2}\big[4K^2\nu_r + 12\nu_r'/t + 12\nu_r/t^2 + c^{(r)}_{\mathrm{Ham}}\big] \quad \text{(algebraic, from (Ham))}, \tag{7}$$

$$w = \tfrac{2i}{K^2}\int_0^t P_\perp c_M^{(r)}\mathrm{d}s, \qquad h = -2\int_0^t \tfrac{s}{t}\tfrac{\sin K(t-s)}{K}\,\mathrm{TT}[c_T^{(r)}](s)\,\mathrm{d}s, \tag{8}$$

$$A_r = (\hat{K}\hat{K} - \tfrac{1}{3}\delta)h_S + (K\otimes w + w\otimes K) + h. \tag{9}$$

At $K=0$ no projector appears: $A_r = -2\int_0^t \frac{s}{t}(t-s)c_T^{(r)}(s)\mathrm{d}s$ and $\nu_r = c_rt^2 - \frac{1}{3}\int_0^t \frac{t^3-s^3}{3ts}c_N^{(r)}\mathrm{d}s$. The lapse particular part is $O(t^4)$; the spatial solution generally starts at $O(t^2)$. The free regular lapse coefficient c_r is fixed by the leading, $O(t^0)$ Hamiltonian constraint, equivalently by the $O(t^2)$ lapse data in (5).

Theorem 5 (Consistency at every order)**.** *The triangular solution satisfies all ten field equations at every order.*

Proof. Let $\mathbb{E}_{\mu\nu} = G_{\mu\nu} - 8\pi G T_{\mu\nu}$. Matter conservation is exact (Lemma 1), so $\nabla_\mu \mathbb{E}^\mu{}_\nu = 0$. Inductively assume that all residuals vanish below order r. At order r this is the background Bianchi identity, with $\mathcal{H} = 1/t$. Set $f = \mathbb{E}^t{}_t$, $v_i = \mathbb{E}^t{}_i$, and $B^j{}_i = \mathbb{E}^j{}_i$, suppressing the order label. Equation (N), which is solved by construction, gives $f = \frac{1}{2}\mathbb{E}$, hence $\operatorname{tr} B = f$. The two Bianchi identities are therefore (at this order B is symmetric, since all lower residuals vanish)

$$f' + \frac{2}{t} f - iK_i v_i = 0, \qquad v_i' + \frac{4}{t} v_i + iK_j B^j{}_i = 0. \tag{10}$$

For $K \neq 0$, the scalar spatial solution imposes (Ham), so $f = 0$; the vector solution imposes $P_\perp v = 0$. The first identity in (10) now gives $K \cdot v = 0$, and thus the *longitudinal* momentum constraint follows rather than being assumed. Consequently $v = 0$, $\operatorname{tr} B = 0$, and the second identity gives $K_j B^j{}_i = 0$. The tensor solution imposes $\mathrm{TT}[B] = 0$. A symmetric tensor that is trace-free and transverse is itself TT, so $B = 0$.

For $K = 0$, the full trace-free spatial equation is solved, and (10) gives $v_i = b_i/t^4$ and $f = b/t^2$. The regular field series has $(M)_i = O(t)$, hence $v_i = O(t^{-1})$; this excludes $b_i \neq 0$. It does *not* exclude $b \neq 0$. Here the initial Hamiltonian constraint is essential: the coefficient c_r from (5) cancels the $O(t^0)$ term of (Ham). Since $\mathbb{E}^t{}_t = -(\mathrm{Ham})/(2N^2)$ and $N \sim a_1 t$, this gives $\lim_{t\to 0} t^2 f = 0$, hence $b = 0$. Finally $\operatorname{tr} B = f = 0$ and $\mathrm{TF}\, B = 0$ give $B = 0$. This completes the induction. □

Remark 6 (Why initial constraint matching cannot be replaced by regularity). At linear order and $K = 0$, take $\nu = \beta t^2$, $H = 0$, $\zeta = 0$ with arbitrary constant β. Then (N), (M) and (T) hold, but $(\mathrm{Ham}) = 36\beta$. The metric variables are even and analytic. This mode is excluded by (5), not by analyticity alone.

5.2 Compact field-source representation

The large component expressions in the ancillary files need not be the working form of the answer. Keep the matrix products and the scalar/vector/tensor projections factored. In particular, put $M = e^H$, $\chi = \zeta + 2\nu$, and let ∇_M and $\mathrm{Ric}[M]$ refer to the spatial metric M. The conformal transformation of spatial Ricci curvature and of the lapse Hessian gives the exact identity

$$\begin{aligned} \mathcal{F} &\equiv \mathrm{TF}\left[R^i{}_j[\gamma] - 2D^i D_j \nu + 2D^i \nu D_j \nu\right] \\ &= e^{-2\zeta}\,\mathrm{TF}\left\{M^{-1}\left[\mathrm{Ric}[M] - \nabla_M \mathrm{d}\chi + \mathrm{d}\chi \otimes \mathrm{d}\chi - 2\mathrm{d}\nu \otimes \mathrm{d}\nu\right]\right\}. \end{aligned} \tag{11}$$

Here $\nabla_M \mathrm{d}\chi$ is the covariant Hessian. Thus (T) is simply $S' + 2(t^{-1} + \nu')S + \mathcal{F} = 0$. Its matrix shear can be generated through cubic order as

$$S = \frac{1}{2}H' + \frac{1}{4}[H', H] + \frac{1}{12}[[H', H], H] + O(\epsilon^4). \tag{12}$$

These commutators must not be dropped: matrices at different orders or momenta need not commute. Together with (N), (M), (Ham) and the Green operators above, these identities give a compact source prescription at arbitrary momenta: retain lower-order fields, set the unknown top-order kernels to zero, and extract the multilinear coefficient. No momenta, endpoint terms or tensor sectors are discarded. The ancillary files contain the expanded coefficients of this factored representation. This is a structural simplification, not a claim that the remaining finite-time integrals collapse to a smaller set of evaluated scalar functions.

5.3 Explicit kernels

First order [closed form].

$$\nu_1 = -(\sin x/x - \cos x), \qquad H_1 = -\tfrac{6}{k^2}\left(\tfrac{2\sin x}{x} - \cos x - 1\right)\left(k_i k_j - \tfrac{1}{3}\delta_{ij}k^2\right), \qquad \zeta_1 = 1.$$

Second order [closed form, general momenta]. Every integral above is of the form $\int_0^t B(s)\mathrm{d}s$, where $B = \sum_\omega e^{i\omega s} R_\omega(s)$ and each R_ω is a Laurent polynomial. The frequencies ω are integer combinations of $p/\sqrt{3}$, $q/\sqrt{3}$, $K/\sqrt{3}$ and K (with $p = |k_1|$, $q = |k_2|$, $K = |k_1 + k_2|$); they vanish on resonance surfaces. We integrate term by term in *entire* functions:

$$\int_0^t s^m e^{i\omega s}\mathrm{d}s = t^{m+1}\mathfrak{e}_m(i\omega t), \qquad \mathfrak{e}_m(z) = \int_0^1 u^m e^{zu}\mathrm{d}u \qquad (m \geq 0), \tag{13}$$

$$\int_0^t s^{-M}\Big[e^{i\omega s} - \sum_{j<M} \tfrac{(i\omega s)^j}{j!}\Big]\mathrm{d}s = J_M(\omega, t), \quad J_1 = -\operatorname{Cin}(\omega t) + i\operatorname{Si}(\omega t), \tag{14}$$

$$J_M = \tfrac{t^{1-M}E_M}{1-M} + \tfrac{i\omega}{M-1}J_{M-1}, \qquad E_M = e^{i\omega t} - \sum_{j<M} \tfrac{(i\omega t)^j}{j!}. \tag{15}$$

Here $\operatorname{Cin}(x) = \int_0^x (1 - \cos u)u^{-1}\mathrm{d}u$ is entire and even, and Si is entire and odd. The subtracted Taylor pieces sum to the principal part of B at $s = 0$, which vanishes (this is asserted in the code). *No division by any frequency occurs*, so the kernels are manifestly finite on every resonance surface:

- the scalar collinear surfaces $\mu = \pm 1$;
- the tensor and vector sound cone $\mu = -1/3$ (for $|k_1| = |k_2|$).

The removable $\ln|\alpha|$ terms that Ci would introduce never appear. The complete expressions for ν_2, h_S, w and h are in `solution/results/second_order_closed_forms.txt`. Each antiderivative is verified by exact differentiation. Substituting the assembled ν_2 back into the lapse equation (N) gives an exact zero in closed form. The scalar part h_S satisfies (Ham) by construction.

The momentum and trace-free equations follow from three facts: the verified antiderivatives, the verified Green functions, and Thm. 5. They are also confirmed independently by the exact series test through $O(t^8)$. A direct closed-form substitution into (M) and (T) is available (`--check all`) but was not run to completion: it takes hours, and it is redundant with these checks.

Third order [explicit nested integrals]. For $C^{(13)}$ only the configuration $(k, q, -q)$ is needed. The source $c^{(3)}$ is explicit (`solution/results/third_order_sources.txt`) in the first-order closed forms and in $K_2(k, q)$, $K_2(k, -q)$ and $K_2(q, -q)$. The last of these is the exact $K = 0$ kernel, never a limit. The third-order kernels are the formulas above applied to $c^{(3)}$, so they are double integrals with closed-form inner integrals.

6 The observable through third order

Proposition 7 (Tetrad)**.** *Fermi–Walker transport along u is parallel transport along ∂_t in $\tilde{g}$, that is, $\mathrm{d}E_a{}^i/\mathrm{d}t = -S^i{}_j E_a{}^j$. With $E(t_I) = \gamma(t_I)^{-1/2}$, the transported frame satisfies $E = \gamma_o^{-1/2} R$ at observation, where $R = 1 + \epsilon^2\Omega/2 + O(\epsilon^3)$ and Ω is antisymmetric. Consequently:*

- *$\Theta^{(1)}$ and $\Theta^{(2)}$ are those of the symmetric frame;*
- *the specified Fermi–Walker map is completed by*

$$\Theta^{(3)}_{\mathrm{FW}} = \Theta^{(3)}_{\mathrm{sym}} + 3(\Omega\hat{n})\cdot\nabla_{S^2}\Theta^{(1)}, \qquad \Omega = \frac{1}{4}\int_{\bar{t}_I}^{\bar{t}_o} [H_1(t,0), H_1'(t,0)]\,\mathrm{d}t; \tag{16}$$

- *$C_\ell^{(22)}$ and $\operatorname{Re} C_\ell^{(13)}$ are individually unchanged.*

Proof. Let $F = \gamma^{-1/2} = e^{-\zeta}e^{-H/2}$. From $E = FR$ and $E' = -SE$, $R' = -F^{-1}(SF + F')R$. Expanding the matrix exponentials gives $R' = \epsilon^2[H_1, H_1']/8 + O(\epsilon^3)$, with $R(t_I) = 1$. This proves the integral in (16); replacing its endpoints by their background values changes R only at $O(\epsilon^3)$, beyond the map order at issue. Since $\Theta_{\rm FW}(\hat{n}) = \Theta_{\rm sym}(R\hat{n})$, the Taylor convention gives the factor 3. The rotation acts on $a^{(1)}_{\ell m}$ by an anti-Hermitian generator, so $\mathrm{Re}\sum_m a^{(1)*}_{\ell m}(D(\Omega)a^{(1)})_{\ell m} = 0$ in every realisation. □

For a monochromatic seed, $H_1(t, 0) = f(t)B$ with a time-independent matrix B, and hence $[H_1, H_1'] = 0$. In that specialisation even the maps agree through third order. The following ray and map formulas use the symmetric frame; (16) completes the map for general spectra after the clock composition.

Take the background ray $\bar{X} = \hat{n}(t_o - t)$, $\bar{e} = -\hat{n}$, and write $X = \bar{X} + \epsilon X_1 + \epsilon^2 X_2/2$. All fields are evaluated on $(t, \bar{X})$, and $\sigma(a, b) = \sigma_{ij}a^i b^j$. Because σ, S and Γ vanish on the background, X_3 and e_3 never enter. The exact consequences of Lemma 2 are as follows.

Integrand.

$$\begin{aligned}
F_1 &= \sigma_{1,nn}, \qquad F_2 = \sigma_{2,nn} + 2X_1\cdot\partial\sigma_{1,nn} - 4\sigma_1(n, e_1),\\
F_3 &= \sigma_{3,nn} + 3X_1\cdot\partial\sigma_{2,nn} - 6\sigma_2(n, e_1) + 3X_2\cdot\partial\sigma_{1,nn} + 3(X_1\cdot\partial)^2\sigma_{1,nn}\\
&\quad - 12(X_1\cdot\partial)\sigma_1(n, e_1) - 6\sigma_1(n, e_2) + 6\sigma_1(e_1, e_1).
\end{aligned}$$

Ray. The displacements are

$$X_r = e_r(t_o)(t - t_o) + \int_t^{t_o}(t' - t)A_r(t')\mathrm{d}t',$$

with

$$\begin{aligned}
A_1 &= 2S_1 n - \Gamma_1(n, n) - n\sigma_{1,nn},\\
A_2 &= [2S_2 n - \Gamma_2(n, n) - n\sigma_{2,nn}] + 2(X_1\cdot\partial)[2S_1 n - \Gamma_1(n, n) - n\sigma_{1,nn}]\\
&\quad + 2[-2S_1 e_1 + 2\Gamma_1(n, e_1) + e_1\sigma_{1,nn} + 2n\sigma_1(n, e_1)].
\end{aligned}$$

The initial directions are the Taylor coefficients of $e(t_o) = -e^{-\zeta}\exp(-H/2)\,\hat{n}$, evaluated at $(t_o, 0)$.

Emission and endpoint. Let φ_r denote ν along the ray, with $\varphi_1 = \nu_1$, $\varphi_2 = \nu_2 + 2X_1\cdot\partial\nu_1$, and so on. Then

$$\begin{aligned}
&d_1 = -\bar{t}_\star\varphi_1, \qquad d_2 = \bar{t}_\star[u_1^2 - \varphi_2 - 2\varphi_1' d_1], \qquad u_1 = d_1/\bar{t}_\star,\\
&I_1 = \int F_1, \qquad I_2 = \int F_2 - 2F_1 d_1, \qquad I_3 = \int F_3 - 3F_1 d_2 - 3F_2 d_1 - 3F_1' d_1^2.
\end{aligned}$$

All unlabelled integrals here run from $\bar{t}_\star$ to t_o; endpoint factors are evaluated at $\bar{t}_\star$. Primes on ray-evaluated quantities denote total derivatives along the background ray.

Normalisation in centred form. At fixed coordinate observation time, put $J_c = I - \langle I\rangle_\Omega$, with coefficients $J_{c,r} = I_r - \langle I_r\rangle_\Omega$, and define $\mathsf{P}_0 f = f - \langle f\rangle_\Omega$. A sky-independent shift of I cancels exactly, so

$$\begin{aligned}
\Theta_c &= \frac{e^{-J_c}}{\langle e^{-J_c}\rangle_\Omega} - 1,\\
\Theta_c^{(1)} &= -J_{c,1},\\
\Theta_c^{(2)} &= -J_{c,2} + \mathsf{P}_0(J_{c,1}^2),\\
\Theta_c^{(3)} &= -J_{c,3} + 3\mathsf{P}_0(J_{c,1}J_{c,2}) - \mathsf{P}_0(J_{c,1}^3) + 3J_{c,1}\langle J_{c,1}^2\rangle_\Omega.
\end{aligned} \tag{17}$$

This is exactly the expanded normalisation formula, with all random-monopole products retained. The dipole is still present inside every product. It is removed only after the complete map is constructed; this final projection does not change $C_{\ell\geq 2}$ and is not a deboost.

Clock. With $\omega(t) = \nu(t,0)$, define separately

$$W_1(T) = \int_0^T t\,\omega_1(t)\,\mathrm{d}t, \qquad W_2(T) = \int_0^T t\,[\omega_2(t) + \omega_1(t)^2]\,\mathrm{d}t.$$

The initial-time shifts d_{I1}, d_{I2} are the emission-root formulas with $\bar{t}_\star \to \bar{t}_I$ and $\varphi \to \omega$. The observation-time shifts are

$$\begin{aligned}
D_1 &= [\bar{t}_I d_{I1} - W_1(\bar{t}_o) + W_1(\bar{t}_I)]/\bar{t}_o,\\
D_2 &= \big[\bar{t}_I d_{I2} + d_{I1}^2 - W_2(\bar{t}_o) - 2\bar{t}_o\omega_1(\bar{t}_o)D_1\\
&\qquad + W_2(\bar{t}_I) + 2\bar{t}_I\omega_1(\bar{t}_I)d_{I1} - D_1^2\big]/\bar{t}_o.
\end{aligned}$$

Compose with $t_o = \bar{t}_o + \epsilon D_1 + \epsilon^2 D_2/2 + \cdots$:

$$\begin{aligned}
\Theta_{\mathrm{sym}}^{(2)} &= \Theta_c^{(2)} + 2D_1\partial_{t_o}\Theta_c^{(1)},\\
\Theta_{\mathrm{sym}}^{(3)} &= \Theta_c^{(3)} + 3D_2\partial_{t_o}\Theta_c^{(1)} + 3D_1\partial_{t_o}\Theta_c^{(2)} + 3D_1^2\partial_{t_o}^2\Theta_c^{(1)}.
\end{aligned}$$

All right-hand sides are evaluated at $t_o = \bar{t}_o$. Equivalently, compose I with this same random time first, centre the resulting exponent, and use (17) once. These are alternative procedures, not two clock corrections to be applied successively.

Fourier dictionary. For a single mode a, with phase $e^{i\lambda k\cdot\hat{n}}$ and $\lambda = t_o - t$, $\mu_a = \hat{k}\cdot\hat{n}$, and $H_{a,ij} = Q_a(t)(k_ik_j - \delta_{ij}k^2/3)$, where $Q_a = -6[2\sin x/x - \cos x - 1]/k^2$:

- $\sigma_{a,nn} = \frac{1}{2}Q_a'k^2(\mu_a^2 - \frac{1}{3})$;
- $\Gamma_a(n,n) = i[2(k\cdot n)n - k] + \frac{i}{2}[2(k\cdot n)H_a n - kH_{a,nn}]$;
- $e_a(t_o) = (1 + \frac{1}{2}H_a(t_o))\hat{n}$;
- $d_a = -\bar{t}_\star\nu_a(\bar{t}_\star)e^{i\lambda_\star k\cdot\hat{n}}$.

The second-order shear is $\sigma_{ab} = \frac{1}{2}H_{ab}' + \frac{1}{4}(H_aH_b + H_bH_a)' + H_a' + H_b'$, and

$$I_{ab} = \int_{\bar{t}_\star}^{t_o}\mathrm{d}t\,\big\{\sigma_{ab,nn} + i(k_b\cdot X_a)\sigma_{b,nn} + i(k_a\cdot X_b)\sigma_{a,nn} - 2\sigma_b(n,e_a) - 2\sigma_a(n,e_b)\big\} - [F_ad_b + F_bd_a]_{\bar{t}_\star},$$
$$\Theta_{ab} = I_aI_b - I_{ab} - \langle I_a\rangle_\Omega I_b - \langle I_b\rangle_\Omega I_a + 2\langle I_a\rangle_\Omega\langle I_b\rangle_\Omega - \langle I_aI_b\rangle_\Omega + \langle I_{ab}\rangle_\Omega + D_a\partial\Theta_b + D_b\partial\Theta_a.$$

A compact expression for *all* map kernels is obtained by taking $\mathcal{J}_a$, $\mathcal{J}_{ab}$ and $\mathcal{J}_{abc}$ to be the centred-exponent kernels *after* clock composition. In the symmetric frame, (17) polarises to

$$\begin{aligned}
\mathcal{K}_1(a) &= -\mathcal{J}_a, \\
\mathcal{K}_2(a,b) &= -\mathcal{J}_{ab} + \mathsf{P}_0(\mathcal{J}_a\mathcal{J}_b), \\
\mathcal{K}_3(a,b,c) &= -\mathcal{J}_{abc} + \mathsf{P}_0\left(\sum_{\text{cyc}} \mathcal{J}_a\mathcal{J}_{bc} - \mathcal{J}_a\mathcal{J}_b\mathcal{J}_c\right) + \sum_{\text{cyc}} \mathcal{J}_a\langle\mathcal{J}_b\mathcal{J}_c\rangle_\Omega.
\end{aligned} \tag{18}$$

Each cyclic sum has three terms. The coefficient convention of §2 introduces no further factorials. For the specified general-spectrum map, add the multilinear coefficient of (16); for the spectra below the symmetric-frame kernels suffice. Thus the lengthy expanded kernels can remain in the ancillary files without obscuring the finite-time structure of the final observable.

Independent generator. `observable_generator.py` rebuilds Θ *from the exact definitions* by fixed-point iteration in the multilinear algebra: the ray ODE, the emission root, the clock equation and the sky normalisation. It uses no hand-expanded formula, and it passes the Ward identities and the superhorizon check of §7 and §9.1.

7 Gauge invariance and Ward identities

Every coordinate and label is a physical scalar or frame: t, X, the surface $T = T_\star$, the fluid line $X = 0$, proper time from $T = T_I$, and $\gamma^{-1/2}$ in the labels. So $\Theta(\hat{n})$ is invariant under any diffeomorphism acting on fields and labels together. The only residual freedom is the conformal group of the primordial chart (§4). Its dilation generator must act as $\zeta \to \zeta + c \Leftrightarrow k \to e^{-c}k$. We verify this exactly:

- *Fields:* $K_2(k,0) = -k\cdot\partial_k K_1$ and $K_3(k,0,0) = (k\cdot\partial_k)^2 K_1$ solve every equation in closed form. Perturbed predictions fail.
- *Observable:* $\Theta_{sL} = -\kappa\partial_\kappa\Theta_s$ and $\Theta_{sLL} = (\kappa\partial_\kappa)^2\Theta_s$ hold through $O(\kappa^4)$, where κ scales the momenta. At this order the ray deflection, time delay, emission shift, clock, frame and normalisation all contribute.
- At first order, the result equals the independent Newtonian-gauge computation (SW, Doppler and ISW) for every ℓ.

7.1 Gradient-mode tests and relation to the literature

Mirbabayi and Zaldarriaga, *CMB Anisotropies from a Gradient Mode* [1], provide two especially useful comparisons. Their §§3–4 show why a physical emission surface, the observer's position and velocity, the observer clock, and the measured temperature normalisation must be retained for a long-gradient cancellation. Their §5 distinguishes an uncompensated primordial gradient, which generates a second-order quadrupole, from a nonlinearly completed adiabatic gradient, which is locally a coordinate transformation. Appendix C demonstrates cancellation of inverse-Laplacian terms in the long-wavelength observable, including boundary contributions.

The symmetry arguments are relevant here, but their numerical transfer coefficients cannot be imported. Their explicit dynamics are matter dominated; the long-long calculation also neglects η_e/η_o, and the squeezed calculation neglects terms of order $(\lambda_S/\lambda_L)^2$. This report instead retains

radiation dynamics and finite emission and observation times. The following checks are derived directly in the temperature frame, without making those matter-era approximations.

Proposition 8 (Exact adiabatic-gradient null test)**.** *Let b be a constant vector, and in a nonsingular patch containing the observer define*

$$D_b(X) = 1 - 2b\cdot X + b^2X^2, \qquad Y_b(X) = \frac{X - bX^2}{D_b(X)}, \qquad \zeta_{\rm ad}(X) = -\ln D_b(X). \tag{19}$$

Then the radiation solution $\nu = H = 0$, $\gamma = D_b^{-2}\delta$ has $\Theta(\hat{n}) = 0$ exactly, at arbitrary finite observation and emission times within the patch.

Proof. The special conformal transformation satisfies $(\partial_X Y_b)^T(\partial_X Y_b) = D_b^{-2}\mathbf{1}$. Thus γ is a flat metric written in different spatial labels, and $g = (a_1 t)^2(-\mathrm{d}t^2 + \mathrm{d}Y_b^2)$ is the unperturbed radiation spacetime. In particular $R_{ij}[\gamma] = 0$, $S = 0$ and all four exact field equations hold. The observer remains at $Y_b(0) = 0$, with $\partial_X Y_b|_0 = \mathbf{1}$, so the specified initial orientation and clock are unchanged. Since $\sigma = 0$, the exact redshift exponent is $I = 0$ for every ray. Normalisation gives $\Theta = 0$, including before dipole subtraction. □

With $b = \epsilon p/2$, the completed primordial profile is

$$\zeta_{\rm ad} = \epsilon\, p\cdot X + \epsilon^2\left[\frac{1}{2}(p\cdot X)^2 - \frac{1}{4}p^2X^2\right] + O(\epsilon^3). \tag{20}$$

The bracket is the coefficient of ϵ^2, not $\zeta^{(2)}$; the latter is twice the bracket in our convention. The completed seed has vanishing curvature, so it also passes (30). An explicit local check is $e^{-\zeta_{\rm ad}} = D_b$, whose Hessian is $2b^2\delta_{ij}$ and has zero trace-free part.

By contrast, for the *uncompleted* local profile $\zeta = \epsilon p\cdot X$ at $X_o = 0$, the leading-gradient formula gives

$$\Theta_{\rm grad,lead}(\hat{n}) = \epsilon^2\tau^2\left[(p\cdot\hat{n})^2 - \frac{1}{3}p^2\right]. \tag{21}$$

It vanishes at first order but not at second order. To compare the primordial completion with [1], use *their* matter-era notation $\phi_1 = q\cdot X = -3\zeta_1/5$, hence $p = -5q/3$. Multiplying the quadratic bracket in (20) by $-3/5$ gives exactly their $\phi_2 = 5q^2X^2/12 - 5(q\cdot X)^2/6$. This agreement concerns the adiabatic initial profile, not equality of the matter and radiation quadrupole coefficients. In particular, adding that quadratic profile would change the Gaussian initial-data prescription of Def. 4; it must not be used to cancel the physical contribution in this report.

Proposition 9 (Soft-gradient identity for the finite-time map)**.** *For the kernels of the complete map at fixed physical labels, assuming differentiability in the soft momentum at zero,*

$$\boxed{\partial_{q_i}\mathcal{K}_2(k,q;\hat{n})|_{q=0} = -\mathcal{D}_i\mathcal{K}_1(k;\hat{n}), \qquad \mathcal{D}_i = k_j\partial_{k_i}\partial_{k_j} - \frac{1}{2}k_i\Delta_k.} \tag{22}$$

Momentum derivatives keep $\hat{n}$, $T_\star$, T_I and τ_o fixed. This identity is valid at general short-mode momentum and finite times; it is not restricted to the superhorizon expansion of the short mode.

Proof. For an arbitrary regular scalar seed ζ_s, its pullback by (19) is the seed $\zeta_s(Y_b(X)) - \ln D_b(X)$. The same observer and tetrad are selected at the origin. Diffeomorphism invariance therefore makes

the complete measured map identical for these two seed descriptions. To first order in an auxiliary parameter λ, set $b = \lambda p/2$ and write

$$Y_b = X + \lambda\xi_p(X) + O(\lambda^2), \qquad \xi_p(X) = (p\cdot X)X - \tfrac{1}{2}X^2 p.$$

For a short seed $sh(X)$ the pulled-back seed is $sh + \lambda p\cdot X + s\lambda\xi_p\cdot\nabla h + \cdots$. If $\mathcal{L}$ and $\mathcal{B}$ denote the linear and symmetric bilinear map responses (with the Taylor prefactor 1/2 on $\mathcal{B}$), equality of the $s\lambda$ coefficients gives

$$\mathcal{B}[h, p\cdot X] = -\mathcal{L}[\xi_p\cdot\nabla h].$$

For $h = e^{ik\cdot X}$, the left-hand side is $-ip_i\partial_{q_i}\mathcal{K}_2|_{q=0}$. The initial deformation on the right is $\xi_p\cdot\nabla e^{ik\cdot X} = -ip_i\mathcal{D}_i e^{ik\cdot X}$, with the displayed k_j coefficients outside the derivatives. Linearity then gives (22). □

Combining this with the dilation identity isolates a soft remainder:

$$\mathcal{R}_2(k,q;\hat{n}) \equiv \mathcal{K}_2(k,q;\hat{n}) + k\cdot\partial_k\mathcal{K}_1(k;\hat{n}) + q_i\mathcal{D}_i\mathcal{K}_1(k;\hat{n}) = O(q^2). \tag{23}$$

Thus both the constant and gradient parts of a soft quadratic response are fixed by the linear transfer function; genuinely new soft information starts at two gradients. This is a useful organisation and consistency condition for the large ancillary kernels. The symmetry proof is exact, and the leading-gradient quadratic kernel independently satisfies (22); a direct arbitrary-time substitution of the full assembled quadratic observable into this identity has not been performed.

What this does and does not cancel. The no-modulation result for a gradient in [1] includes the appropriate transformation of short-mode initial data (or the corresponding inflationary consistency condition). An exactly Gaussian g with an arbitrary spectrum in a fixed marked chart does not by itself impose that condition. Equations (22) and (23) therefore do not license dropping the soft pieces of $\mathcal{K}_2$, or the gradient-squared term in (21), from this problem's spectrum. Nor do long-wavelength cancellations prove that the kernels at comparable, finite momenta reduce to local curvature polynomials. In the notation of §9.1, the positive m_2^2 contribution is retained.

8 The spectrum

8.1 Gaussian reduction

With $\mathcal{K}_r(k_1..k_r;\hat{n})$ the kernel of $\Theta^{(r)}$ (observer at the origin) and $A^{(r)}_{\ell m} = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{K}_r$:

$$C_\ell^{(11)} = \tfrac{1}{2\ell+1}\textstyle\sum_m\int_k P|A^{(1)}_{\ell m}|^2, \qquad C_\ell^{(22)} = \tfrac{2}{2\ell+1}\textstyle\sum_m\int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2,$$
$$C_\ell^{(13)} = \tfrac{3}{2\ell+1}\textstyle\sum_m\int_k\int_q P(k)P(q)A^{(1)}_{\ell m}(k)A^{(3)}_{\ell m}(k,q,-q)^*.$$

The disconnected part of 22 is proportional to $\delta_{\ell 0}$. In 13 all three pairings survive; the internal $(q,-q)$ pairing is a pairing of the fourth moment and is not removed by normal ordering. Odd orders vanish.

Compact finite-time spectrum. For $\ell \geq 2$, define the corrected linear response

$$A^{\mathrm{eff}}_{\ell m}(k) = A^{(1)}_{\ell m}(k) + \frac{\epsilon^2}{2}\int_q P(q)A^{(3)}_{\ell m}(k,q,-q). \tag{24}$$

Then the complete result through fourth order can be written as two norms:

$$\boxed{C_\ell = \frac{1}{2\ell+1}\sum_m \left\{ \epsilon^2 \int_k P(k)|A^{\rm eff}_{\ell m}(k)|^2 + \frac{\epsilon^4}{2}\int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2 \right\} + O(\epsilon^6).} \tag{25}$$

The square is understood only through $O(\epsilon^4)$: its $O(\epsilon^6)$ part is not a prediction. To see the factor $1/2$ in (24), Wick-decompose the cubic Gaussian product into its normal-ordered part plus its three single contractions. The cubic prefactor $1/6$ turns these into a $1/2$ correction to the linear response; the remaining normal-ordered cubic part is orthogonal to the linear part. The quadratic contribution has two connected pairings and its prefactor is $(1/2)^2$, giving the other $1/2$ in (25). No 13 interference term has been discarded or absorbed into a redefinition of the primordial variance. The formula applies whenever the displayed integrals exist, in particular to the monochromatic case.

Equations (18)–(25), together with the factored field sources in §5.2, are a compact organisation of the *general finite-time* answer. They retain the same integrals as the expanded result; they do not establish an additional cancellation of the cubic response or evaluate its momentum and time integrals.

Linear transfer function. For $\ell \geq 2$, the convention is

$$\begin{aligned} A^{(1)}_{\ell m}(k) &= 4\pi i^\ell T_\ell(k) Y^*_{\ell m}(\hat{k}), \\ T_\ell(k) &= -3\int_{x_\star}^{x_o} s'(x)\, w_\ell\Big(\sqrt{3}(x_o - x)\Big)\, \mathrm{d}x, \\ s(x) &= \frac{2\sin x}{x} - \cos x, \qquad w_\ell(y) = \frac{1}{3} j_\ell(y) + j''_\ell(y). \end{aligned} \tag{26}$$

The prime on s differentiates x, and those on j_ℓ differentiate its full argument. Thus $C_\ell^{(11)} = 4\pi\int \mathcal{P}(k) T_\ell(k)^2\, \mathrm{d}\ln k$. Evaluated expressions for T_2 and T_3 remain in `verification/results/T2_closed_form.txt` and `verification/results/T3_closed_form.txt`.

8.2 Monochromatic spectrum

After the complete observable is constructed, set

$$\mathcal{P}(k) = \sigma^2 \delta(\ln k/k_0), \qquad P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\delta(k-k_0), \qquad \langle g^2\rangle = \sigma^2 \text{ held fixed.}$$

The background and the labels $T_\star$, T_I and τ_o are also held fixed. Using joint rotation invariance, with $\hat{k} = \hat{z}$ and $\hat{q}_\mu = (\sqrt{1-\mu^2}, 0, \mu)$:

$$C_\ell^{(11)} = 4\pi\sigma^2 T_\ell(k_0)^2, \tag{27}$$

$$C_\ell^{(22)} = \frac{\sigma^4}{4\pi}\int_{-1}^{1}\mathrm{d}\mu \int \mathrm{d}\Omega \int \mathrm{d}\Omega'\, P_\ell(\hat{n}\cdot\hat{n}')\, \mathcal{K}_2(\hat{n})\mathcal{K}_2(\hat{n}')^*, \tag{28}$$

$$C_\ell^{(13)} = \frac{3\sigma^4}{2} i^\ell T_\ell(k_0) \int_{-1}^{1}\mathrm{d}\mu \Big[\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n})\Big]^*. \tag{29}$$

All sky integrals in closed form. Every term of $\mathcal{K}_r$ has the form $c(\lambda,\dots)\,\Pi(\hat{n})\,e^{i\hat{n}\cdot V}$. Here $V=\sum_a k_a\lambda_a$ collects the line-of-sight and nested times, and Π is a polynomial. By Funk–Hecke and $n_i\to -i\partial_{V_i}$:

$$\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\,\Pi(\hat{n})e^{i\hat{n}\cdot V}=\Pi(-i\partial_V)\big[4\pi i^\ell j_\ell(|V|)P_\ell(\hat{z}\cdot\hat{V})\big],$$

$$\iint P_\ell(\hat{n}\cdot\hat{n}')\Pi(\hat{n})e^{i\hat{n}\cdot V}\Pi'(\hat{n}')e^{-i\hat{n}'\cdot V'}=\Pi(-i\partial_V)\Pi'(i\partial_{V'})\big[(4\pi)^2 j_\ell(|V|)j_\ell(|V'|)P_\ell(\hat{V}\cdot\hat{V}')\big].$$

The sky *integrals* at $\ell=0$ give $4\pi j_0$; the sky *means* in the normalisation terms include the factor $1/(4\pi)$ and therefore give j_0. The remaining integrals are therefore:

- for $C^{(22)}$: μ, plus per copy one line-of-sight time, one ray time and one field-kernel time;
- for $C^{(13)}$: μ and up to five nested times.

All integrands are elementary or closed-form.

Theorem 10 (Existence)**.** *For all finite $0<x_\star<x_o$, the integrals for $C_\ell^{(22)}$ and $C_\ell^{(13)}$ converge absolutely. The restriction to the shell is well defined.*

Proof. The domains are compact. The integrands are bounded for the following reasons:

- the field kernels are continuous in time and momenta, and finite on resonances, by the entire-function representation;
- $c_M^{(2)}$ is analytic and vanishes at $K=0$, so $c_M=O(K)$, and the projector terms $(\hat{K}\hat{K}-\delta/3)h_S$ and $K\otimes w$ are bounded near $K\to 0$. They are discontinuous only at $\mu=-1$, a null set;
- $K_2(q,-q)$ in $C^{(13)}$ is the exact $K=0$ kernel;
- the ray integrals are over finite intervals.

□

9 Limits

9.1 Superhorizon limit, exact in ζ

Theorem 11. *Let all momenta scale as $\kappa\to 0$ at fixed times. Then, to all orders in ϵ,*

$$\Theta(\hat{n})=-\tau^2\,e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta-\partial_i\zeta\partial_j\zeta\right]_o\hat{n}^i\hat{n}^j+O(\kappa^3),\qquad \tau^2=\tfrac{1}{6}(\bar{t}_o^2-\bar{t}_\star^2).$$

Proof. At zeroth gradient order the local seed is constant, $\zeta=\zeta_o$, and the ray direction is $e=-e^{-\zeta_o}\hat{n}$, not $-\hat{n}$. Spatial variation of ζ changes this direction by $O(\kappa)$, while H and σ are $O(\kappa^2)$. Their spatial variation along the finite ray adds one further gradient. Since $e^{-2\zeta_o}\sigma=\frac{1}{2}(e^H)'$ locally, the constant conformal factor cancels against the two direction vectors, giving

$$I_{\mathrm{lead}}=\frac{1}{2}n^in^j[H_{ij}(\bar{t}_o,0)-H_{ij}(\bar{t}_\star,0)].$$

Endpoint and clock shifts are $O(\kappa^2)$ and multiply $O(\kappa^2)$ quantities. The Fermi–Walker correction also enters beyond the displayed order. Substitution of (5) gives the claimed I_{lead}. It is trace-free, so $\langle I_{\mathrm{lead}}\rangle_\Omega=0$ and $\Theta=-I_{\mathrm{lead}}+O(\kappa^3)$.

At each fixed amplitude order the remainder is uniform in the angles on compact momentum and time domains, so the gradient expansion may be taken inside the compact integrals. This statement does not exchange an infinite amplitude series with the full nonlinear solution or its ensemble average.

The independent generator confirms the formula at first and second order, at (k, q) with general momenta. The corresponding third-order run at $(k, q, -q)$ was *not completed* (§10). At third order the formula rests on two things: the proof above, which uses only the gradient-expansion data (5) (exact in ζ and verified for arbitrary ζ), and the generator's third-order Ward identity W3. □

Geometric form. Let $f = e^{-\zeta}$ and $\gamma^{(0)} = e^{2\zeta}\delta$ be the primordial seed metric. Since $f\,\partial_i\partial_j f = -e^{-2\zeta}(\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta)$, the theorem has the equivalent forms

$$\boxed{\Theta_{\rm lead}(\hat{n}) = \tau^2 f(\partial_i\partial_j f)_{\rm TF} n^i n^j = \tau^2 R_{\langle ab\rangle}[\gamma^{(0)}] n^a n^b.} \tag{30}$$

Here $R_{\langle ab\rangle}$ is the trace-free spatial Ricci tensor in the orthonormal frame $e^{-\zeta}\partial_a$ of the *seed* metric. It is not the Ricci tensor of the physical metric at the observation event. The leading observable is a local curvature quadrupole.

Proposition 12 (Gaussian average of the leading-gradient map)**.** *Let $m_j = \int k^j \mathcal{P}(k)\,\mathrm{d}\ln k$ for $j = 0, 2, 4$, with these moments finite. For $\zeta = \epsilon g$, the exact Gaussian average of the leading-gradient expression is*

$$\boxed{C_{2,\rm lead} = \frac{16\pi}{225}\tau^4\epsilon^2 e^{8\epsilon^2 m_0}\left(m_4 + \frac{5}{3}\epsilon^2 m_2^2\right).} \tag{31}$$

For a general spectrum this is the leading-gradient coefficient when all momenta scale to zero at fixed times, not a formula for finite-wavelength modes.

Proof. Put $A_{ij} = (\partial_i\partial_j g)_{\rm TF}$ and $B_{ij} = (\partial_i g\,\partial_j g)_{\rm TF}$. The leading quadrupole tensor is $Q = -\tau^2 e^{-2\epsilon g}(\epsilon A - \epsilon^2 B)$. Isotropy and Gaussianity imply that g, A and ∇g are mutually independent: the g–Hessian covariance is pure trace and all odd-derivative covariances vanish. Thus g is independent of B and

$$\langle A : A\rangle = \frac{2}{3}m_4, \qquad \langle B : B\rangle = \frac{10}{9}m_2^2, \qquad \langle A : B\rangle = 0, \qquad \langle e^{-4\epsilon g}\rangle = e^{8\epsilon^2 m_0}.$$

The ensemble mean of Q vanishes. For a trace-free tensor Q, $\sum_m |a_{2m}|^2 = \frac{8\pi}{15}Q : Q$, so $C_2 = \frac{8\pi}{75}\langle Q : Q\rangle$, proving (31). □

On the monochromatic shell, $m_0 = \sigma^2$, $m_2 = k_0^2\sigma^2$ and $m_4 = k_0^4\sigma^2$. With $\Xi = x_o^2 - x_\star^2 = k_0^2(\vec{t}_o^{\,2} - \vec{t}_\star^{\,2})/3$,

$$\boxed{C_{2,\rm lead} = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2 e^{8\epsilon^2\sigma^2}\left(1 + \frac{5}{3}\epsilon^2\sigma^2\right).} \tag{32}$$

This resums the Gaussian average of the leading-gradient map only. It is not a convergence claim for the full amplitude expansion and supplies no arbitrary-$k_0 t_o$ spectrum. Its expansion, $e^{8v}(1+5v/3) = 1 + 29v/3 + O(v^2)$ with $v = \epsilon^2\sigma^2$, reproduces the fourth-order answer. Separately, exact Wick moments give the leading $O(k_0^4)$ coefficients:

$$C_2^{(11)} = \frac{4\pi}{225}\sigma^2\Xi^2, \qquad C_2^{(22)} = \frac{272\pi}{675}\sigma^4\Xi^2, \qquad C_2^{(13)} = \frac{16\pi}{75}\sigma^4\Xi^2,$$

$$C_2 = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2\Big[1+\frac{29}{3}\epsilon^2\sigma^2\Big] + O(\epsilon^6) + O(k_0^6), \qquad C_{\ell\geq 3} = O(k_0^6).$$

Two checks: $C_2^{(11)}$ agrees with the line-of-sight transfer function $T_2 \to -\frac{2}{15}k_0^2\tau^2$, and the direct ϵ-expansion of $\langle\sum_m |a_{2m}|^2\rangle$ reproduces $\frac{1}{4}C^{(22)} + \frac{1}{3}C^{(13)}$. The fourth-order correction is a finite, positive renormalisation of the quadrupole by the factor $1+\frac{29}{3}\epsilon^2\sigma^2$. It is independent of T_I at this order.

9.2 Late time, first order

Proposition 13. *For $\ell \geq 2$ and fixed $x_\star$, $T_\ell(x_\star, x_o) = O(\ln x_o/x_o)$ as $x_o \to \infty$. Hence $C_\ell^{(11)} \to 0$.*

Proof. Write $T_\ell = -3\int_0^{x_o-x_\star} s'(x_o-u)w_\ell(\sqrt{3}u)\mathrm{d}u$, with $w_\ell = j_\ell/3 + j_\ell''$. Two integrations by parts give $\int_0^\infty e^{iu}w_\ell(\sqrt{3}u)\mathrm{d}u = 0$ exactly for $\ell \geq 2$, because $j_\ell(0) = j_\ell'(0) = 0$. This is the observer-local sound-cone cancellation. Since $s'(x) = \sin x + O(1/x)$, the $O(1)$ oscillation therefore vanishes. The remaining pieces are oscillatory with distinct frequencies $\sqrt{3}\pm 1$ and amplitudes $O(1/x_o)$, giving $O(\ln x_o/x_o)$ by integration by parts. □

At fourth order the late-time limit is non-uniform at the resonances $\mu = \pm 1$ and $\mu = -1/3$. Only the order "angle first" is admissible. Its evaluation is [open].

9.3 General spectra

Theorem 14 (Infrared)**.** *For spectra smooth on $(0,\infty)$, consider the $q \to 0$ region. By the Ward identities, $K_2(k,q) = -k\cdot\partial_k K_1(k) + O(q)$ and $K_3(k,q,-q) = (k\cdot\partial_k)^2 K_1(k) + O(q^2)$. Here the kernels are analytic in q at fixed finite times, and the absence of an $O(q)$ term in K_3 follows from its symmetry under $q \to -q$. The $O(q)$ cross term in $|K_2|^2$ is odd in $\hat{q}$ and integrates to zero.*

Both the q-long and the k-long regions of $C^{(22)}$ contribute. Let $\sigma_L^2 = \int_{q<\Lambda} P$ and $D = k\cdot\partial_k$. The infrared-sensitive part of $\frac{1}{4}C^{(22)} + \frac{1}{3}\operatorname{Re}C^{(13)}$ is then

$$\frac{\sigma_L^2}{2\ell+1}\int_k P(k)\Big[\sum_m |DA_{\ell m}^{(1)}|^2 + \operatorname{Re}\sum_m A_{\ell m}^{(1)}D^2A_{\ell m}^{(1)*}\Big]$$
$$= \frac{\sigma_L^2}{2(2\ell+1)}\int_k \Big[\sum_m |A_{\ell m}^{(1)}|^2\Big](D+3)^2P(k).$$

The equality uses two integrations by parts in $\ln k$ and assumes vanishing radial boundary terms; for spectra with explicit cutoffs those boundary terms must be retained. Under that assumption the expression vanishes for scale-invariant $P \propto k^{-3}$. Otherwise it is the physical rescaling of the short-mode spectrum by the long-mode variance. The remainder is $O(q^2)P(q)$, which is integrable for $\mathcal{P}(q)$ with $\int_0 \mathcal{P}(q)q^2\,\mathrm{d}\ln q < \infty$.

Ultraviolet [power counting only]. At fixed finite times:

- the first-order kernels are bounded uniformly in k;
- each order of the sources carries at most two spatial derivatives;
- each ray displacement carries one.

Counting powers on this basis gives $|\mathcal{K}_2(k,q)| \lesssim q^2$ and $|\mathcal{K}_3(k,q,-q)| \lesssim q^4$ as $q \to \infty$. That suggests absolute convergence of $C^{(22)}$ and $C^{(13)}$ whenever $\int^\infty \mathcal{P}(q) q^4 \,\mathrm{d}\ln q < \infty$.

This is *not* a proof. A rigorous bound needs uniform estimates of the lapse Green function near $s \to 0$ (where $|G_N|$ grows like $1/(K^3 t s)$) against sources whose small-s Taylor coefficients grow with q. The true large-q behaviour for a scale-invariant spectrum, including possible cancellations, is [open].

10 Verification ledger

All checks are exact symbolic computations (SymPy 1.14). None uses floating point or quadrature as evidence. "Generic 2-jets" means that every function value and derivative is an independent symbol, so a zero residual is a proof.

Check	Script and scope
Lemmas L1–L5; generic field equations	`verification/check_exact_lemmas.py`, `check_fw_conformal.py`, `check_field_equations_generic.py` (generic 2-jets)
Mixed trace-free form (T)	`solution/check_evolution_form.py` (generic 2-jets)
First order, two gauges, all ℓ	`verification/check_first_order.py`; `solution/first_order_engine_check.py`
Closed form T_2, T_3	`verification/transfer_closed_form.py`
Green functions, $K \to 0$	`solution/check_green.py`
2nd-order sources and operators	`solution/second_order.py`
2nd order, all 10 equations	`solution/second_order_consistency.py`: exact Taylor series through $O(t^8)$, general K and $K = 0$
2nd-order closed forms	`solution/closed_second_order.py`: every antiderivative by exact differentiation; (N) by closed-form substitution (`--check N`); (M) and (T) via the theorem and the series test
3rd-order sources and operators	`solution/third_order.py`
3rd order, all 10 equations	`solution/fast_series.py 8`: the pair kernels, including the exact $K = 0$ pair, then the triple at $(k, q, -q)$ with general momenta. The *full nonlinear* equations are recomputed with the solved kernels and vanish through $O(t^6)$. The arithmetic is exact (sparse polynomials over $\mathbb{Q}(i)$ with formal inverse momenta; denominators are cleared exactly at the zero test). Negative controls (a wrong c_3, a corrupted $K = 0$ pair kernel) fail at the expected orders. The run takes about 70 s. `third_order_consistency.py` is the slower SymPy reference implementation of the same test.
Dilation Ward identities (fields)	`solution/check_ward_dilation.py`, `check_ward_dilation3.py` (closed form; negative controls)

Check	Script and scope
Observable expansion formulas	`solution/check_observable_expansion.py` (arbitrary fields, general jets)
Observable Ward identities	`solution/observable_generator.py W2|W3` through $O(\kappa^4)$
Superhorizon observable	`solution/observable_generator.py SH`: orders 1 and 2, general (k, q). *Not completed:* `SH3` (third order at $(k, q, -q)$). Its fields are built by the fast engine in about 12 s, but the generator's SymPy assembly of the observable did not finish within 50 minutes. Porting the generator to the fast arithmetic would remove this; it was not done.
Superhorizon C_2	`solution/superhorizon.py`
Strengthened report identities	`solution/check_report_identities.py`: background residual Bianchi identities and zero-mode constraint matching; conformal driver and cubic commutator shear; Fermi–Walker rotation; centred normalisation and polarised map kernels; the two-norm spectrum; general-moment and monochromatic leading-gradient resummation. These are exact symbolic identities, not an arbitrary-time cubic-observable verification.
Literature-inspired gradient tests	`solution/check_gradient_modes.py`: exact special-conformal Jacobian and flat seed; zero shear and hence zero map for the completed adiabatic profile; quadratic completion and its conversion to the convention of [1]; nonzero uncompensated-gradient quadrupole; plane-wave soft generator and the leading-gradient check of (22). The general finite-time Ward identity is proved by covariance, not by full-kernel substitution.
Normalisation; Wick and Fourier measures	`verification/normalization_expansion.py`; `validation/check_algebra.py`

11 What remains unevaluated

1. For general k_0, $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are given as exact absolutely convergent integrals: over μ and the nested times, with closed-form integrands, and with all sky integrals evaluated. The compact source, map and two-norm representations retain these same integrals. No further general cancellation or smaller evaluated closed form has been established; this is not a proof that none exists. The integrals are *not* evaluated beyond the superhorizon limit. The expanded ancillary equations remain available as a reference rather than being embedded in the report. The literature-inspired (23) fixes the first two soft coefficients but does not determine the comparable-momentum remainder or the full cubic response.

2. The late-time limit at fourth order (non-uniform at the resonances) is not evaluated.

3. Ultraviolet convergence for spectra without compact support is established by power counting only, not by a rigorous bound.

4. The infrared statement (Thm. 14) assumes that the kernels are analytic in q at $q = 0$ at fixed finite times. This holds for the integral representations by construction, but the uniformity of the expansion in k over unbounded spectra is not proved.

5. The perturbative domain is treated order by order: a single emission intersection and no caustics. There is no statement about convergence in ϵ. In particular, (32) is an exact average of a leading-gradient expression, not a justification for interchanging the full nonlinear solution, an infinite amplitude series and an ensemble average.

References

[1] M. Mirbabayi and M. Zaldarriaga, *CMB Anisotropies from a Gradient Mode*, arXiv:1409.4777, version 2 (2015). The supplied source is `arXiv-1409.4777v2/gradient_submit_Apr11_15.tex`; the comparisons above use §§3–5 and Appendix C.